

# X-ray Spectroscopic Diagnostics of Confined and Eruptive Flares

## Using SoLEXS/Aditya-L1

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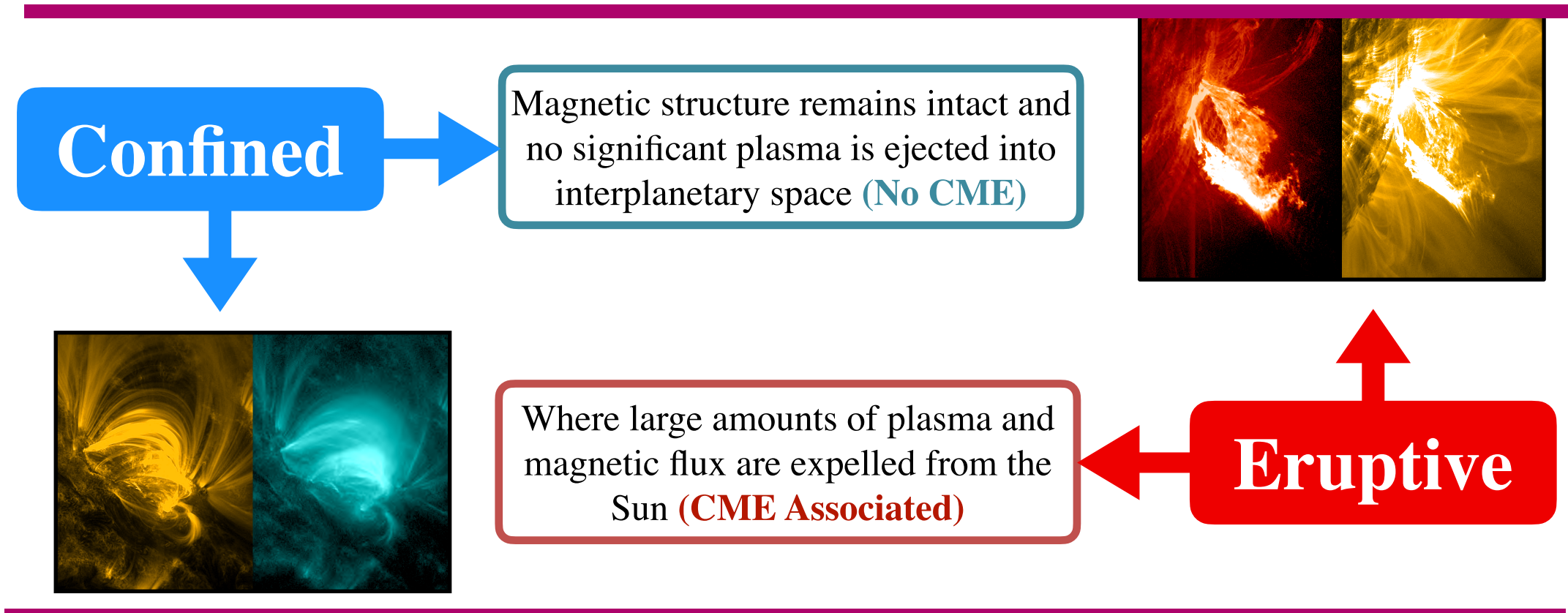


### Abstract

Solar flares are broadly classified as confined (no CME) or eruptive (with CME). This study performs a comparative X-ray analysis to quantify the distinct physical characteristics that govern a flare's eruptive potential. We analyze high-cadence soft X-ray spectra of 43 flares observed by the Solar Low Energy X-ray Spectrometer (SoLEXS) onboard Aditya-L1. By fitting the spectra, we derive the temporal evolution of key plasma parameters, including temperature (T), emission measure (EM), and elemental abundances. Our results reveal a clear dichotomy: confined flares are systematically hotter than their eruptive counterparts, while eruptive flares exhibit a significantly higher emission measure, indicating a larger volume of heated plasma. These findings provide direct observational evidence of the different thermodynamic conditions that distinguish confined flares from large-scale eruptive events.

### INTRODUCTION

- A solar flare is a **sudden, intense release of radiation** caused by **magnetic energy** release in the Sun's atmosphere.
- Flares generally occur near **sunspots in active regions**, where twisted magnetic field lines reconnect and release energy.
- Flares can be classified as
  1. Confined
  2. Eruptive



| Flare Class | Associated X-ray Flux (W/m <sup>2</sup> ) |
|-------------|-------------------------------------------|
| A           | < 10 <sup>-7</sup>                        |
| B           | 10 <sup>-7</sup> - 10 <sup>-6</sup>       |
| C           | 10 <sup>-6</sup> - 10 <sup>-5</sup>       |
| M           | 10 <sup>-5</sup> - 10 <sup>-4</sup>       |
| X           | > 10 <sup>-4</sup>                        |

### SOLEXS & DATA PRODUCTS

- The **Solar Low Energy X-ray Spectrometer (SoLEXS)** is a soft X-ray spectrometer aboard Aditya-L1, to measure the solar soft X-ray flux to study flares.
- Sun as a Star Spectrometer** (intergrated flux)

#### Data Products -

- Good Time Intervals Files:** Time periods of valid data collection by the detector
- Light Curve Files:** 1-second-cadence count rates (2–22 keV)
- Pulse Invariant Files:** Calibrated energy spectra recorded every second.

| Parameter                             | Specification                     |
|---------------------------------------|-----------------------------------|
| Energy Range                          | 2–22 keV                          |
| Energy Resolution                     | (2.2–22 keV for spectral fitting) |
| Time Cadence                          | 1 second                          |
| Spectral Channel                      | 1 second                          |
| Temporal Channel                      | 1 second                          |
| Detector                              | Silicon Drift Detector (SDD)      |
| Type                                  | 2 (named SDD1 & SDD2)             |
| Active Area                           | 400 ± 30 mm <sup>2</sup>          |
| Thickness                             | 8 µm thick Be                     |
| Entrance Window                       | SDD: 7.106 mm <sup>2</sup>        |
| Aperture Area                         | SDD: 0.106 mm <sup>2</sup>        |
| Field of View                         | SDD: ±1.8°                        |
| Calibration Source                    | Fe-55 with Ti foil                |
| Digital Pulse Processing parameters   | Spectral Channel: 2 µs            |
| Pulse peaking time (triangular pulse) | Temporal Channel: 0.35 µs         |
| Number of channels in the spectrum    | 340                               |
| Channel Widths                        | 1–168 channel: ~47.6 eV           |
|                                       | 169–340 channel: ~95.2 eV         |

### EVENT SELECTION

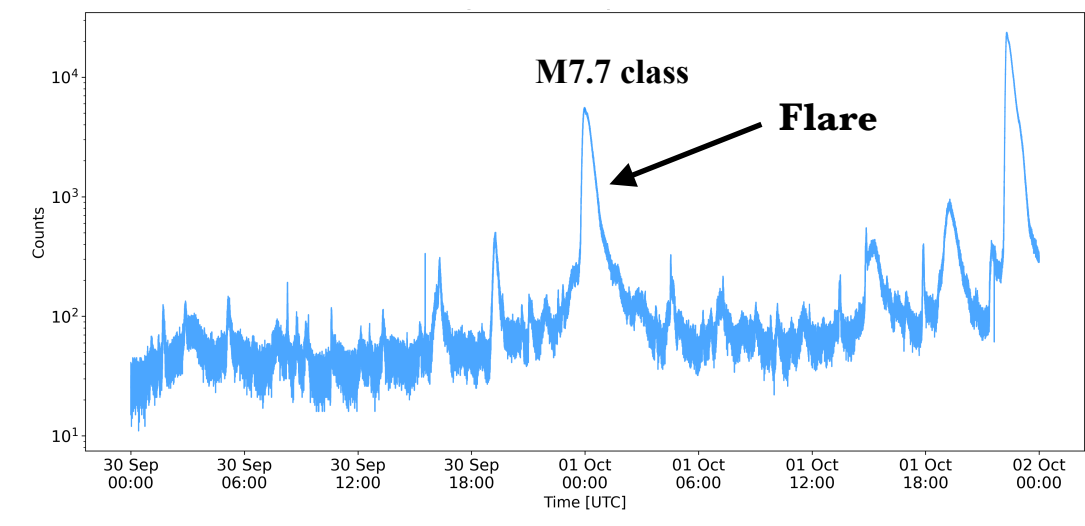
- Flare identification**
  - Selected events from the **GOES Flare Catalog**
  - Time period: **1 July 2024 – 31 July 2025**
- Verification**
  - Cross-checked with **SoLEXS X-ray light curves**
- CME association**
  - Identified using the **SOHO/LASCO CME catalog**

| Flare Class | Eruptive | Confined | Total |
|-------------|----------|----------|-------|
| C-class     | 4        | 4        | 8     |
| M-class     | 17       | 7        | 24    |
| X-class     | 10       | 1        | 11    |
| Total       | 30       | 13       | 43    |

### METHODOLOGY

#### Data Processing

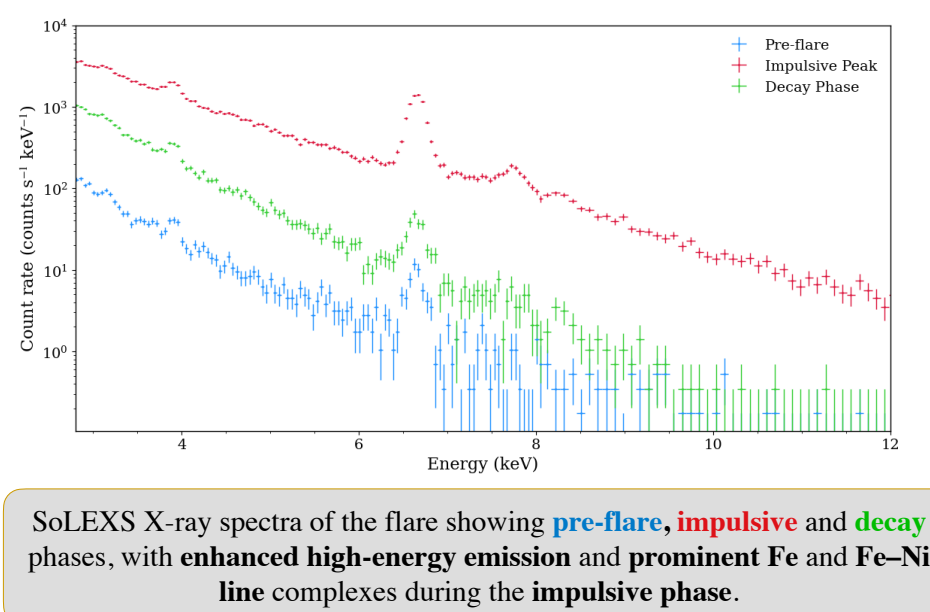
Identify the **Flare** in the **Solexs lightcurve**



#### Spectra Generation

**Spectral Files** were generated for each Flare.

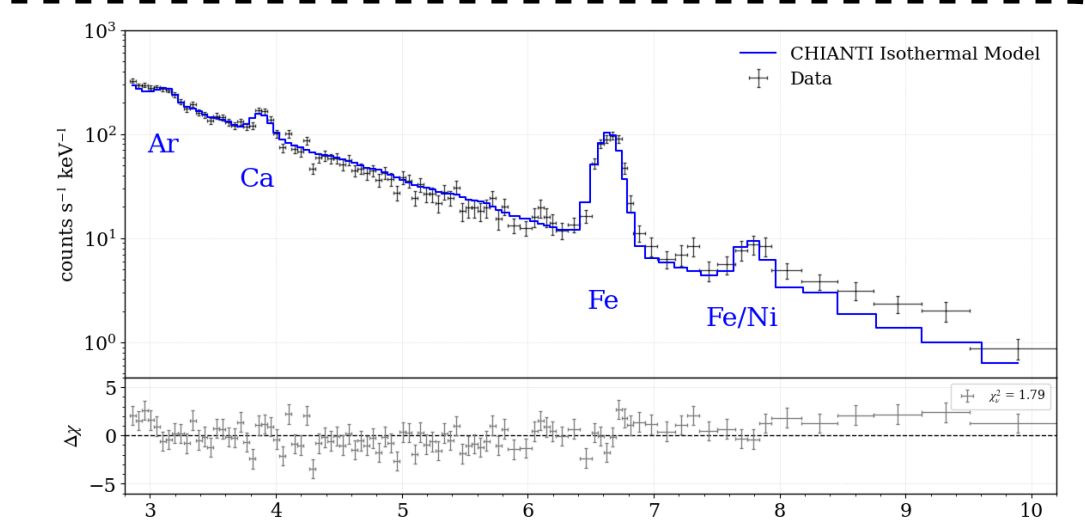
Time bin intervals were optimized to ensure a high **signal-to-noise ratio** and **goodness of fit**.



#### Model & Free Paramters

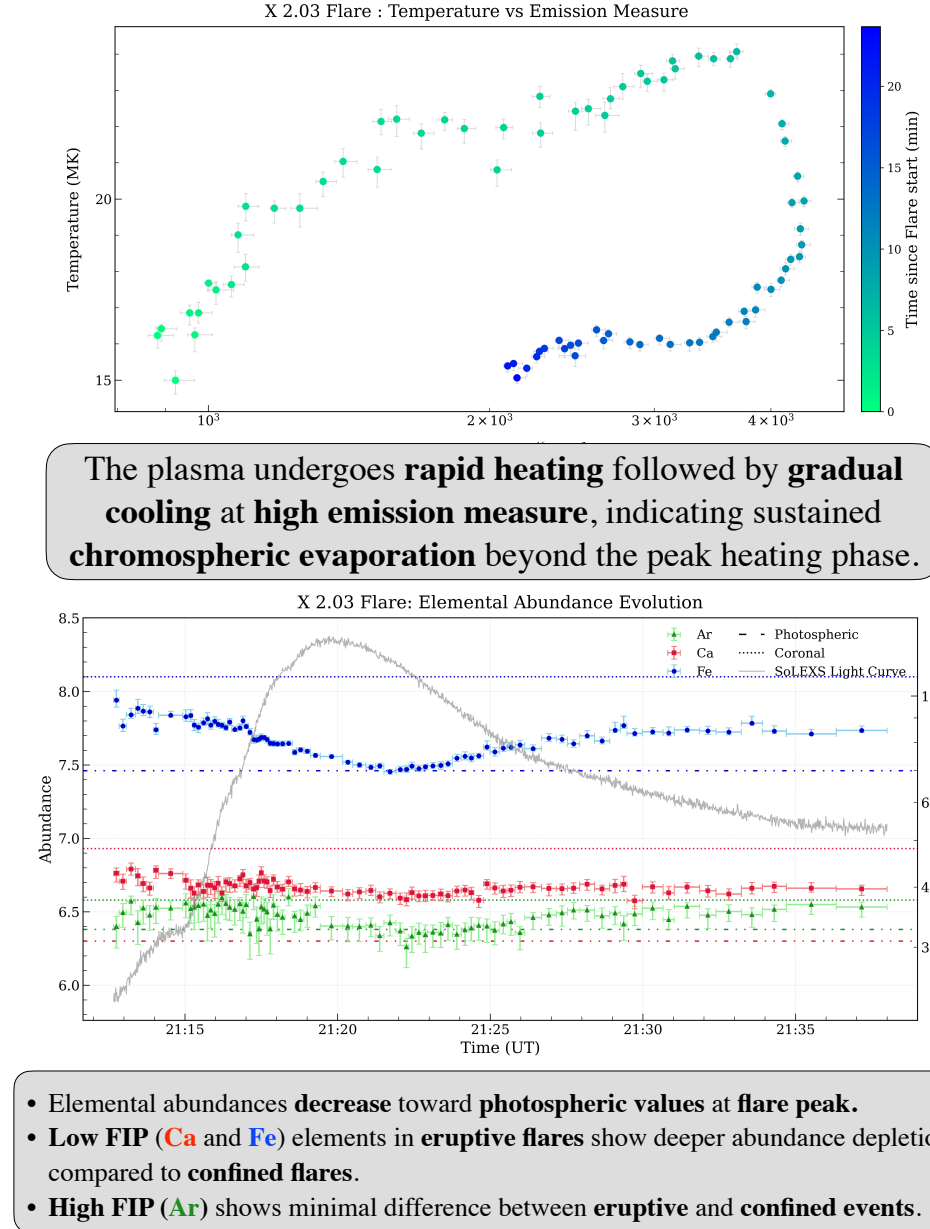
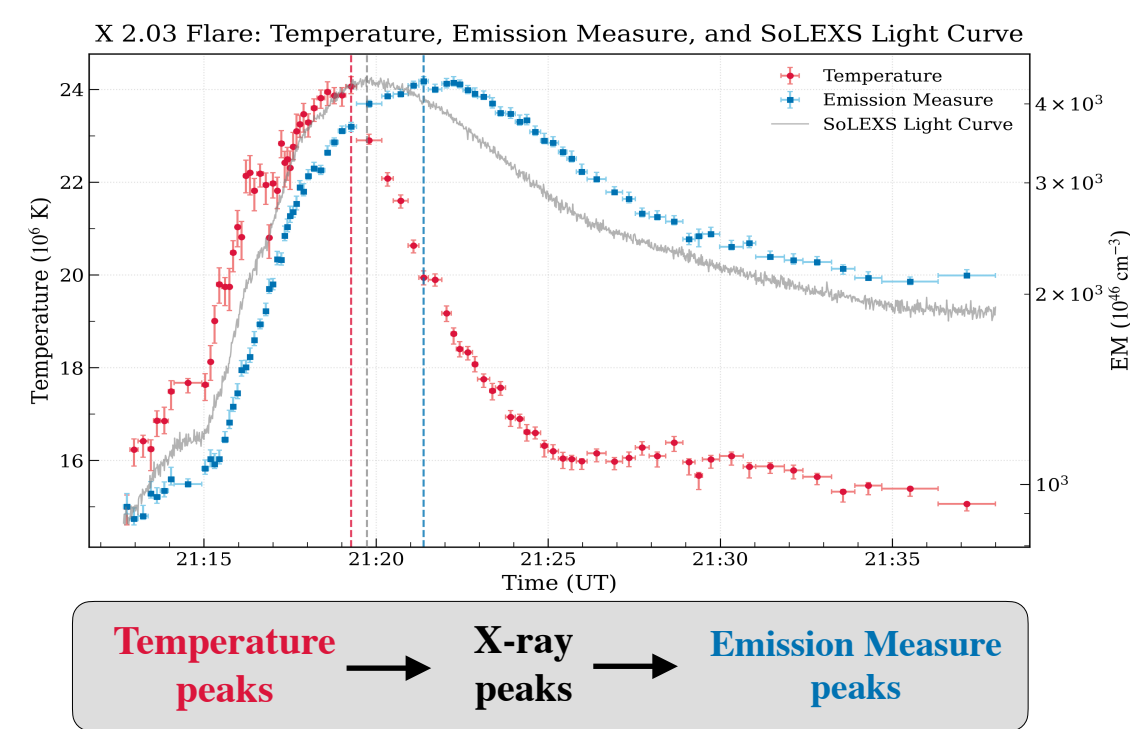
Spectra were fitted with **PyXSPEC** using the **Chianti Isothermal model** (chisoth).

Energy range used: **2.8–12 keV**

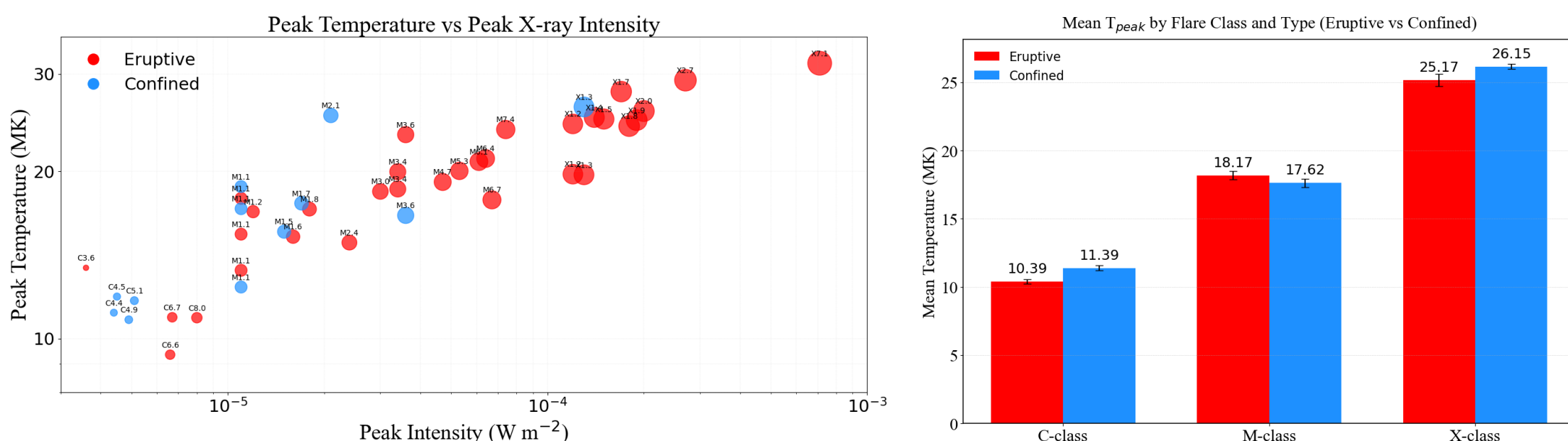


#### Spectral Fitting

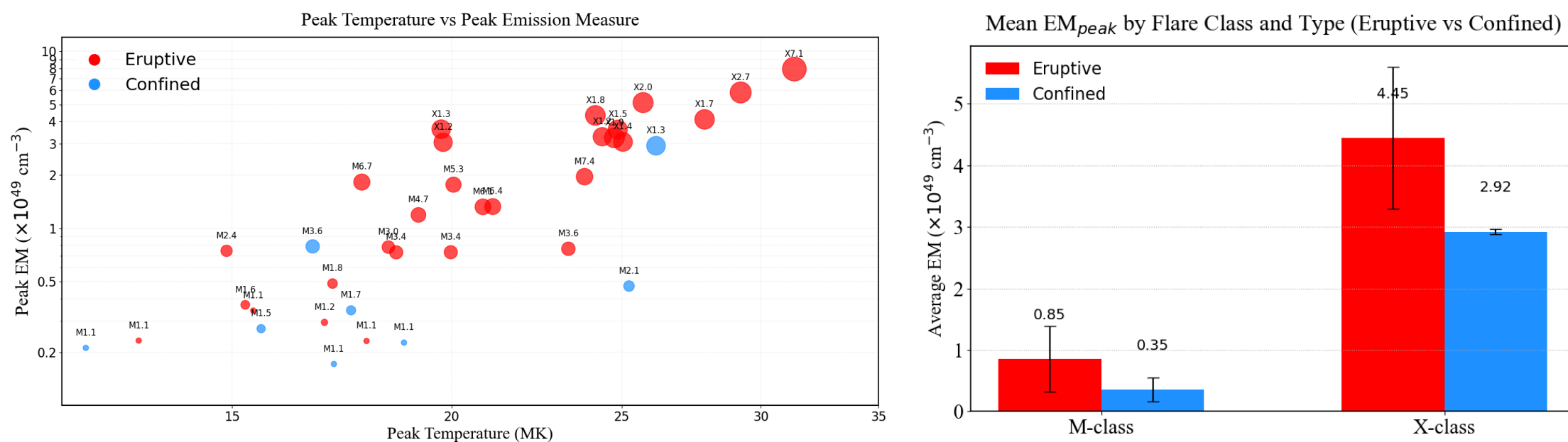
**Temperature, Emission Measure, and Abundances of Ar, Ca, Fe, Ni** are left free during fitting



### RESULTS



- C-Class & X-Class:** **Confined flares** show **higher peak temperatures** than **Eruptive**.
- M-class flares:** **Mixed behavior** — some **confined flares** are hotter, others not.



- Peak Emission Measure **increases** with Peak Intensity (Flare Class).
- Eruptive Flares** have significantly **higher Emission Measure** than **Confined Flares**.

### CONCLUSION

- Significant **Time lag** between Peak Temperature, Peak X-ray Flux, and Peak Emission Measure.
- Peak Temperature and Peak Emission Measure **increase** with Peak Intensity.
- Confined Flares may exhibit **higher peak temperatures** than Eruptive Flares
- Eruptive Flares have a **higher EM** than Confined Flares
- Confined Flares show a **more photospheric abundances** of low FIP elements during flare peak.

### REFERENCES

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- XSPEC: The First Ten Years.** Astronomical Data Analysis Software and Systems.(Arnaud, K.A., 1996.)
- Evolution of Elemental Abundances during B-Class Solar Flares: Soft X-Ray Spectral Measurements with Chandrayaan-2 XSM.** (Mondal, B., Vadavale V. S., 2021.)
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- Statistical relationship between solar flares and coronal mass ejections.** (S. Yashiro and N. Gopalswamy, 2009)
- A Database of Magnetic and Thermodynamic Properties of Confined and Eruptive Solar Flares.** (M. D. Kazachenko, 2023)

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